



OPEN ACCESS

EDITED BY

Sonsoles López-Pernas,
University of Eastern Finland, Finland

REVIEWED BY

Nick Baker,
University of Windsor, Canada

*CORRESPONDENCE

Hong Wang
✉ goohongzi@jlu.edu.cn

RECEIVED 30 November 2025

REVISED 02 April 2026

ACCEPTED 06 April 2026

PUBLISHED 22 April 2026

CITATION

Wang H and Shan W (2026) The safety gap: restoring productive struggle through pedagogically aligned generative AI.
Front. Educ. 11:1757622.
doi: 10.3389/feduc.2026.1757622

COPYRIGHT

© 2026 Wang and Shan. This is an open-access article distributed under the terms of the [Creative Commons Attribution License \(CC BY\)](#). The use, distribution or reproduction in other forums is permitted, provided the original author(s) and the copyright owner(s) are credited and that the original publication in this journal is cited, in accordance with accepted academic practice. No use, distribution or reproduction is permitted which does not comply with these terms.

The safety gap: restoring productive struggle through pedagogically aligned generative AI

Hong Wang* and Wenhui Shan

Cancer Center, The First Hospital of Jilin University, Changchun, Jilin, China

The integration of Generative Artificial Intelligence (GenAI) into education presents a paradox: the more helpful the tool, the more it risks eroding the cognitive processes essential for deep learning. Educational research has long shown that meaningful learning depends not only on correct answers, but also on productive struggle, schema construction, and forms of scaffolding that preserve rather than bypass learner effort. This challenge is especially acute for novice learners, whose understanding is still developing and for whom premature answer delivery may replace the generative processes through which durable knowledge is formed. This paper introduces the concept of the “Safety Gap”—the divergence between a student’s AI-assisted performance and their internal capability to verify that output, leading to potential epistemic risks when the tool fails. While GenAI offers unprecedented opportunities for personalized support and content generation, current Oracle-style models often eliminate the productive struggle necessary for schema construction. Drawing on recent literature, we argue that educational AI should be designed not merely to maximize convenience or answer completion, but to support cognitive engagement and independent judgment. We therefore propose a pedagogical framework centered on Socratic and Adversarial AI architectures that withhold direct solutions, introduce constructive cognitive friction, and prioritize process-based assessment over final outputs.

KEYWORDS

cognitive offloading, generative artificial intelligence, medical education, productive struggle, safety gap

1 Introduction: the paradox of the helpful machine

The integration of Generative Artificial Intelligence (GenAI) into educational settings is not merely a technological upgrade; it represents a fundamental restructuring of the cognitive contract between learner and knowledge. From our perspective in medical education, where clinical competency is non-negotiable, this shift presents a critical juncture. On one side lies the promise of the super-powered student, augmented by an omnipresent tutor capable of synthesizing vast ontologies in milliseconds. Conversely, this integration introduces a subtle challenge often obscured in standard metrics: the Safety Gap.

We define the Safety Gap as the widening chasm between the surface-level competence a student displays when supported by helpful AI tools and their actual, unassisted cognitive capability (Gerlich, 2025). Here, unassisted capability refers to the

learner's independently reproducible reasoning without AI support at the moment of task performance. It is the difference between a polished diagnosis or essay submitted by a student—generated by a Large Language Model (LLM)—and the fragmented reasoning they can summon when the screen goes dark (Figure 1).

This figure is intended as a conjectural conceptual model rather than an empirical plot. It illustrates a critical divergence between visible output and internal understanding. While Augmented Expertise (Line B) represents the desirable outcome of AI enhancing already solid human capability, the Safety Gap (red shaded area) emerges when novices achieve high surface performance (Line C) via frictionless AI without developing the corresponding internal capability (Line A). Here, Line A refers to the learner's independently reproducible reasoning without AI support, Line B to the desirable augmentation of established human competence, and Line C to AI-assisted output quality.

The “Safety Gap” is now defined not as high performance itself, but as the “Epistemic Discrepancy”—the inability of the user (Line A) to verify the output of the tool (Line C).

Current dominant design paradigms prioritizing speed, fluency, and frictionless assistance are fundamentally misaligned with the neurological requirements of deep learning (Jose et al., 2025a). As Rus and Kendeou (2025) argue, unrestrained access to LLMs is at odds with effective learning because these tools often eliminate productive struggle, the very process necessary for reflection, analysis, and synthesis (Rus and Kendeou, 2025). This paper argues that to close the Safety Gap, we must stop asking how AI can make schoolwork easier, and start asking how AI can make it harder in the right ways.

2 The cognitive crisis: offloading and automation bias

To understand the danger of the Safety Gap, we must examine how frictionless GenAI interacts with the cognitive architecture of learning.

2.1 Cognitive offloading and schema construction

Learning involves the construction of schemas, mental structures that organize knowledge and support transfer. In educational terms, this process depends not only on exposure to correct answers, but also on productive struggle through which learners test, revise, and consolidate understanding. For novice learners in particular, schema formation depends on engaging in the generative work required to interpret, connect, and apply new ideas. Struggle with a complex concept is therefore not merely an obstacle to learning, but part of the process through which knowledge structures are formed and reorganized. By contrast, when an LLM provides an immediate polished explanation, the learner may evaluate the output without fully engaging in the processes required for durable understanding.

While cognitive offloading can reduce immediate task demands, it risks bypassing the desirable difficulties necessary for deep learning. As Kapur demonstrates through the

framework of *Productive Failure*, engaging with complex problems without immediate assistance can lead to better conceptual understanding (Kapur, 2014, 2016). Furthermore, Sweller's (1988) *Cognitive Load Theory* suggests that while managing intrinsic load is crucial, eliminating the germane load required for schema construction is counterproductive (Sweller, 1988).

Cognitive offloading is not inherently detrimental. For advanced learners with well-established schemas, AI may reduce routine processing demands and preserve working memory for higher-order judgment. For novices, however, the same convenience may short-circuit the generative effort through which foundational competencies are built.

2.2 Automation bias and the “fluency heuristic”

The risk is compounded by automation bias, the tendency to favor output generated by automated systems. LLMs speak with absolute confidence and linguistic fluency, triggering a heuristic in the human brain: “If it sounds this coherent, it must be true” (Gerlich, 2025).

In high-stakes fields like medicine or engineering, this is critical. A student may intellectually know an AI can hallucinate, but cognitively, they often treat it as an objective Oracle. This leads to a surrender of epistemic agency, where the learner stops feeling responsible for verifying the truth of the output. If a generation of learners loses this agency, we risk an epistemic collapse where knowledge becomes a recursive loop of AI-generated content unchecked by human critical thought.

3 The architectural landscape: from oracles to tutors

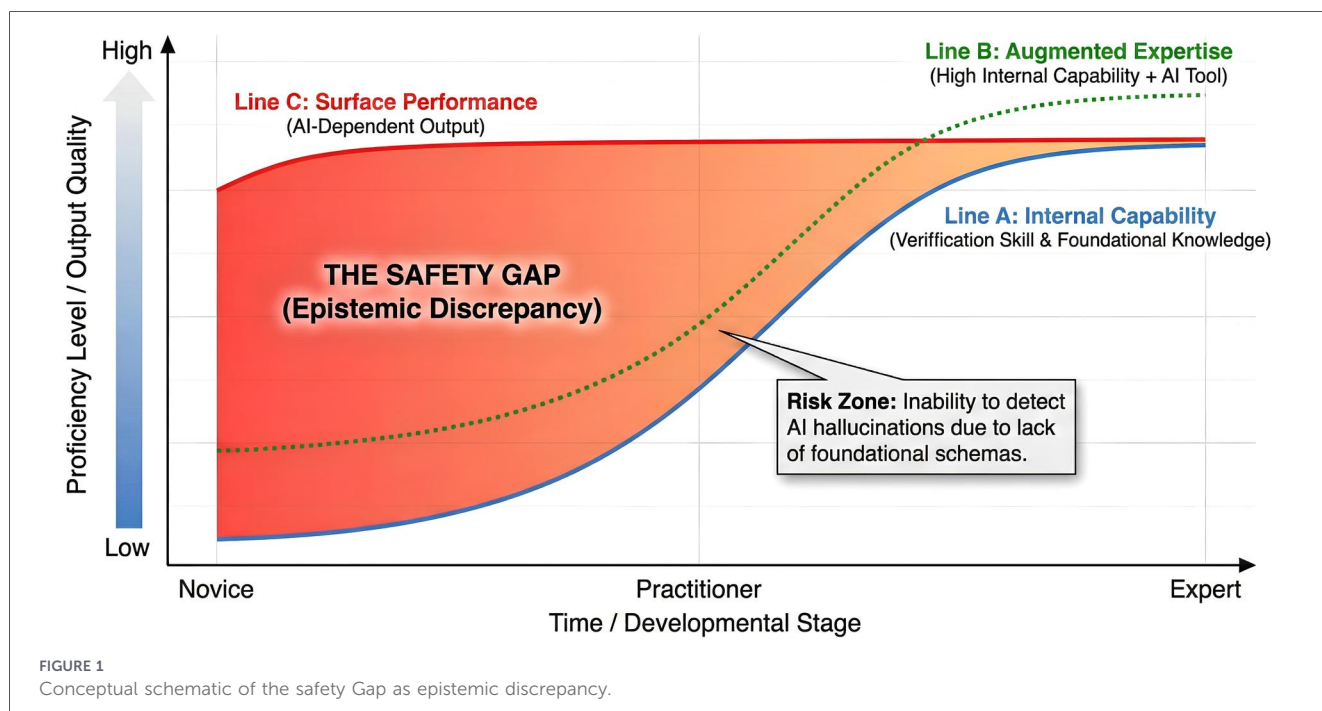
Not all AI is created equal. The extent of the Safety Gap depends heavily on the specific design and personality of the educational software. Here, “Oracle” is used as our own analytic label for answer-delivery systems optimized for immediacy, rather than as terminology taken from any single cited source. We categorize the current landscape into three architectures:

3.1 The oracle (generic LLMs)

Typified by standard chatbots (e.g., ChatGPT/Gemini/Grok), these models are trained to be helpful, often defined as satisfying the user's request immediately. While excellent for information retrieval, they are the primary drivers of the Safety Gap because they maximize performance but minimize learning (Allen et al., 2026).

3.2 The socratic tutor

A growing subset of tools designed specifically for education. These agents are typically configured via specific system prompts or educational wrappers to withhold direct answers



(Khakpaki, 2025). As described by Wang et al. (2024), Confusion Helper applications do not provide solutions but generate pedagogical guidance or hints to help students solve problems themselves (Wang et al., 2024). By reintroducing friction, these tools support schema construction rather than replacing it.

3.3 The adversarial agent

An experimental category where the AI is programmed to play Devil's Advocate, critiquing the student's work or presenting plausible errors ("lures") that the student must identify. This forces the student into a defensive posture, requiring high-germane load critical thinking to win the argument (Jose et al., 2025b). A comparison of these architectural approaches and their implications for the Safety Gap is presented in Table 1.

4 Bridging the gap: A pedagogical framework

To harness GenAI without deskilling students, we propose a four-part strategy: *Restrict, Realign, Require, and Refine*.

4.1 Restrict: creating cognitive sanctuaries

Educational settings should preserve some AI-free spaces to ensure that baseline human capability can still be observed, practiced, and assessed. Just as physical dissection is required to understand anatomy, mental dissection—unassisted reasoning sessions—must be preserved. Rus and Kendeou (2025) advocate for an inverted scaffolding approach: beginning with highly restricted access that prioritizes learner effort, and progressively increasing access to AI tools only as learners demonstrate domain mastery and self-regulation (Rus and Kendeou, 2025). For example, in a clinical reasoning session, students could be required to submit an initial differential diagnosis without AI assistance before being allowed to consult an AI tool for comparison and revision.

4.2 Realign: implementing Socratic architectures

While major AI providers may not restrict their general models, educational institutions can deploy wrapper applications driven by specific system prompts that govern the tool's behavior. Educational AI tools should follow a withholding

TABLE 1 A pedagogical Spectrum of generative AI architectures in education.

Tool category	Design philosophy	Cognitive function	Risk of safety gap	Example
The oracle	Helpfulness/speed	Information retrieval (answer delivery)	High (eliminates struggle)	Generic ChatGPT/Gemini/Grok
The socratic tutor	Guided discovery	Scaffolding/schema construction	Low (supports struggle)	SocraticLM/Khanmigo
The adversarial agent	Critique/friction	Epistemic vigilance (defense)	Minimal (builds resilience)	Devil's advocate bots

This table categorizes AI tools based on their interaction design, contrasting generic Oracle models that prioritize immediacy with education-specific architectures (Socratic and Adversarial) designed to induce productive struggle and mitigate the Safety Gap.

protocol that delays direct answer delivery for conceptual or diagnostic tasks. If a query is purely factual (e.g., What is the dosage of X?), the AI may act as an Oracle. However, if the query is conceptual or diagnostic, the system must switch to Socratic Mode, offering hints in decreasing order of abstraction rather than immediate answers. For instance, if a student asks a conceptual question, the system could first request the learner's own explanation, then provide a hint, and only later reveal a fuller answer if needed.

Recent reviews of GenAI in learning analytics indicate that while current implementations often focus on automated classification or scoring, there is immense potential for human-AI collaboration that facilitates “guided discovery” rather than simple answer delivery.

4.3 Require: process-based assessment

In the age of GenAI, grading the final product (the essay or the code) is insufficient. We must shift to *Process-Based Assessment*. Recognizing that reviewing every interaction log is not scalable, we advocate for assessing metacognitive reflection. Students should be required to submit a brief rationale explaining *why* they accepted or revised the AI's suggestions, thereby evidencing their critical engagement. A passing grade should require evidence that the student critically evaluated the AI's suggestions, justified revisions, and verified the final output, rather than blindly accepting the machine's response (Thesen and Park, 2025). In practice, this could take the form of a short reflection in which students explain one AI suggestion they accepted, one they rejected, and how they verified the final answer (Zimmerman, 2002).

4.4 Refine: adversarial learning

GenAI can also be used to create opportunities for productive failure and critical evaluation. Instead of artificially engineering misinformation, educators should leverage the inherent limitations of current LLMs—such as cultural biases, hallucinations in complex medical cases, or confident but incorrect citations—as organic adversarial examples for students to critique (Golchini et al., 2025). Finding and refuting AI hallucinations turns the tool's weakness into a critical thinking exercise, transforming the interaction from passive consumption to active evaluation. For example, students could be asked to identify a plausible but unsupported claim in an AI-generated explanation and then revise the answer using verified sources.

5 Discussion and future directions

The integration of GenAI into the Learning Analytics (LA) cycle offers profound opportunities. As Misiejuk et al. (2025) highlight in their systematic review, GenAI is currently being used to automate discourse coding, generate synthetic data for training, and develop sophisticated dashboards (Misiejuk et al., 2025). However,

they also note that evidence on the effectiveness of GenAI on actual learning is still lacking.

To ensure these tools support rather than hinder learning, future educational use should prioritize pedagogically aligned deployments rather than unrestricted answer-delivery systems. Rather than simply training models on vast web corpora, developers and educators must adapt models using Retrieval-Augmented Generation (RAG) or pedagogically aligned system prompts that prioritize desirable difficulties.

The Safety Gap is not a reason to reject the technology, but a mandate to master it. The true value of a human in the age of AI lies not in retrieving facts—the machine wins that race—but in metacognition, empathy, and epistemic agency (Estaphan et al., 2025). It is our duty to design systems that challenge, confound, and stretch students, ensuring that learners retain independent judgment, metacognitive awareness, and the capacity to verify AI-assisted outputs.

Data availability statement

The original contributions presented in the study are included in the article/Supplementary Material, further inquiries can be directed to the corresponding author.

Author contributions

HW: Writing – review & editing, Funding acquisition, Writing – original draft. WS: Visualization, Writing – review & editing, Resources.

Funding

The author(s) declared that financial support was received for this work and/or its publication. Jilin University Undergraduate Education Reform Research Project, 2023XZD098, to HW Jilin Provincial Higher Education Reform Project, 20250PSFX6P002Y, to HW.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was not used in the creation of this manuscript.

Any alternative text (alt text) provided alongside figures in this article has been generated by Frontiers with the support of artificial intelligence and reasonable efforts have been made to ensure accuracy, including review by the authors wherever possible. If you identify any issues, please contact us.

Publisher's note

All claims expressed in this article are solely those of the authors and do not necessarily represent those of their affiliated

organizations, or those of the publisher, the editors and the reviewers. Any product that may be evaluated in this article, or claim that may be made by its manufacturer, is not guaranteed or endorsed by the publisher.

References

- Allen, J. W., Levy, N., and Wilkinson, D. (2026). Empowering patient autonomy: the role of large language models (LLMs) in scaffolding informed consent in medical practice. *Bioethics* 40 (2), 183–193. doi: 10.1111/bioe.70030
- Estaphan, S., Kramer, D., and Witchel, H. J. (2025). Navigating the frontier of AI-assisted student assignments: challenges, skills, and solutions. *Adv. Physiol. Educ.* 49 (3), 633–639. doi: 10.1152/advan.00253.2024
- Gerlich, M. (2025). AI tools in society: impacts on cognitive offloading and the future of critical thinking. *Societies* 15 (1), 6. doi: 10.3390/soc15010006
- Gerlich, M. (2025). From offloading to engagement: an experimental study on structured prompting and critical reasoning with generative AI. *Data* 10 (11), 172. doi: 10.3390/data10110172
- Golchini, NB, Passalacqua, E, Vaughn, L, Abdunnour, R-EE, Zack, T, and Finlayson, SG. Socratic AI: An Adaptive Tutor for Clinical Case Based Learning. *medRxiv [Preprint]*. 2025. doi: 10.1101/2025.06.22.25329661
- Jose, B., Cherian, J., Verghis, A. M., Varghise, S. M., S. M., and Joseph, S. (2025a). The cognitive paradox of AI in education: between enhancement and erosion. *Front. Psychol.* 16, 1550621. doi: 10.3389/fpsyg.2025.1550621
- Jose, B., Cleetus, A., Joseph, B., Joseph, L., Jose, B., and John, A. K. (2025b). Epistemic authority and generative AI in learning spaces: rethinking knowledge in the algorithmic age. *Front. Educ.* 10:1647687. doi: 10.3389/feduc.2025.1647687
- Kapur, M. (2014). Productive failure in learning math. *Cogn. Sci.* 38 (5), 1008–1022. doi: 10.1111/cogs.12107
- Kapur, M. (2016). Examining productive failure, productive success, unproductive failure, and unproductive success in learning. *Educ. Psychol.* 51 (2), 289–299. doi: 10.1080/00461520.2016.1155457
- Khakpaki, A. (2025). Advancements in artificial intelligence transforming medical education: a comprehensive overview. *Med. Educ. Online* 30 (1), 2542807. doi: 10.1080/10872981.2025.2542807
- Misiejuk, K., López-Pernas, S., Kaliisa, R., and Saqr, M. (2025). Mapping the landscape of generative artificial intelligence in learning analytics. *J. Learn. Anal.* 12 (1), 12–31. doi: 10.18608/jla.2025.8591
- Rus, V., and Kendeou, P. (2025). Are LLMs actually good for learning? *AI Soc.* 40 (7), 5617–5618. doi: 10.1007/s00146-025-02323-9
- Sweller, J. (1988). Cognitive load during problem solving: effects on learning. *Cogn. Sci.* 12 (2), 257–285. doi: 10.1207/s15516709cog1202_4
- Thesen, T., and Park, S. H. (2025). A generative AI teaching assistant for personalized learning in medical education. *NPJ. Digit. Med.* 8 (1), 627. doi: 10.1038/s41746-025-02022-1
- Wang, S., Xu, T., Li, H., Zhang, C., Liang, J., Tang, J., et al. Large Language Models for Education: A Survey and Outlook. *arXiv [Preprint]*. 2024. arXiv:2403.18105v2.
- Zimmerman, B. J. (2002). Becoming a self-regulated learner: an overview. *Theory Pract.* 41 (2), 64–70. doi: 10.1207/s15430421tip4102_2